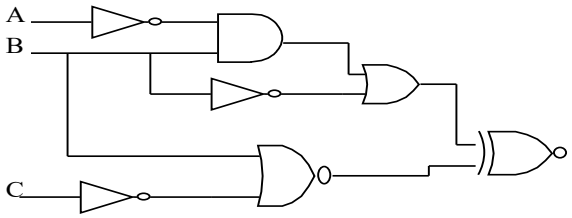
Digital Electronics – Senior Division

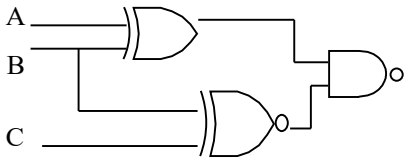
1. **00-01 C3 Digital Electronics**

   Find all ordered triples that make the following circuit TRUE.



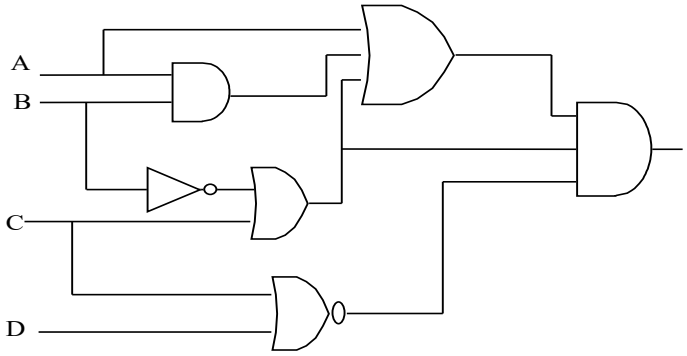
2. **00-01 C3 Digital Electronics**

   Write a Boolean Algebra expression that is an exact translation of the following circuit. Do not simplify.



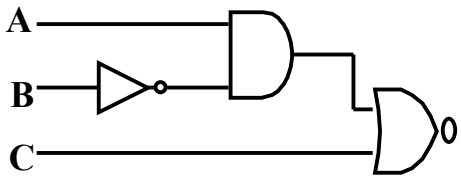
3. **00-01 C4 Digital Electronics**

   An AND gate with three inputs is TRUE only when all three inputs are TRUE and an OR gate with three inputs is FALSE only when all three inputs are FALSE. How many ordered quadruples make the following TRUE?

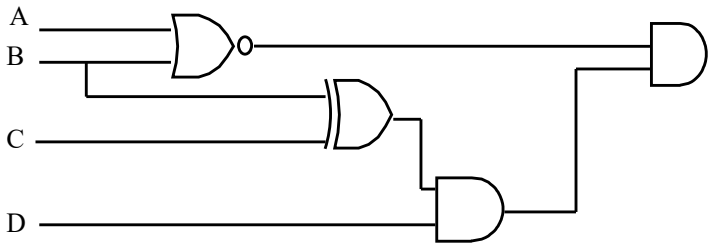


4. **01-02 C3 Digital Electronics**

   Find all ordered triples (A,B,C) that make the circuit TRUE.

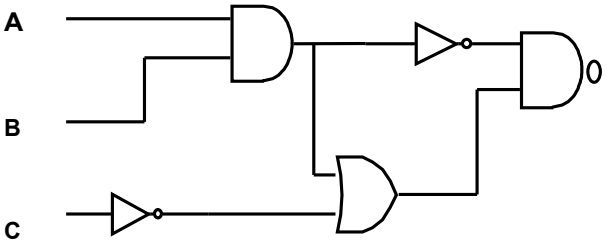


5. **01-02 C3 Digital Electronics**
Suppose that B is always FALSE. Find a simplified Boolean expression for the circuit.



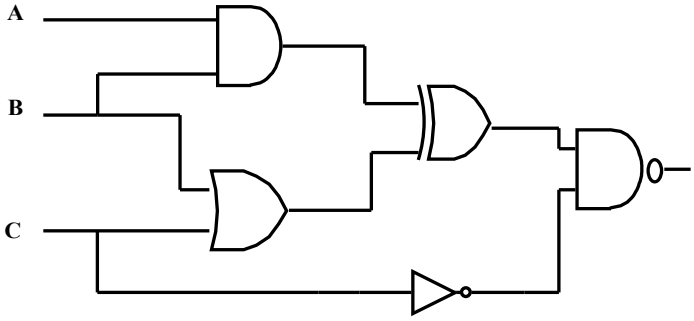
6. **01-02 C4 Digital Electronics**

Find all ordered triplets (A,B,C) that make the following circuit TRUE.



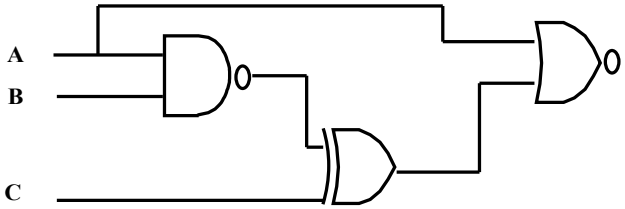
7. **02-03 C3 Digital Electronics**

Find all ordered triples that make the following circuit FALSE:



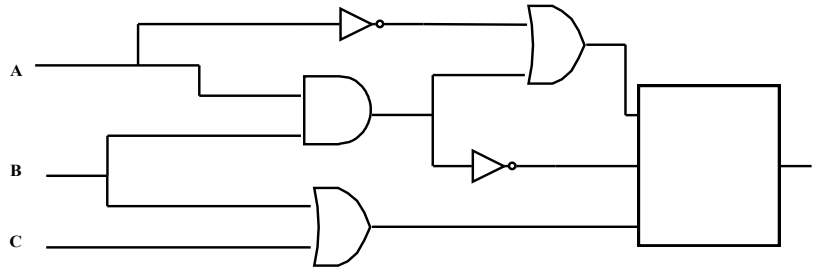
8. **02-03 C3 Digital Electronics**

How many ordered triples make the circuit



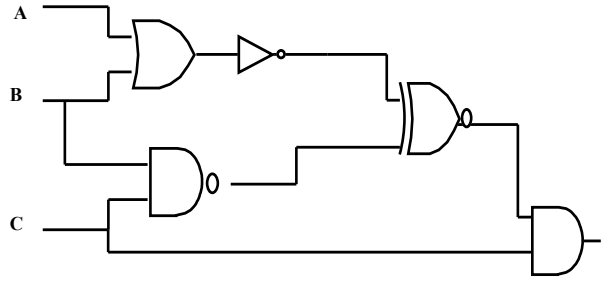
9. **02-03 C4 Digital Electronics**

The logic gate  is True if all of its inputs are either all True or all False.
It is False otherwise.
Find all ordered triples that make the following circuit True.



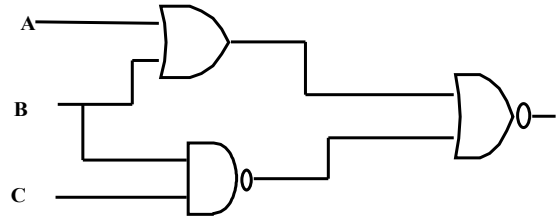
10. **03-04 C3 Digital Electronics**

Construct a truth table with 8 columns – make columns for A, B and C and the results of the 5 logic gates. How many entries are FALSE?



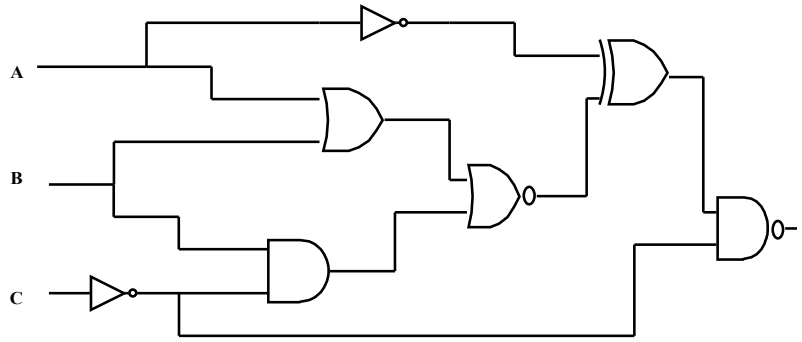
11. **03-04 C3 Digital Electronics**

Find all ordered triples (A,B,C) that make the following circuit TRUE.



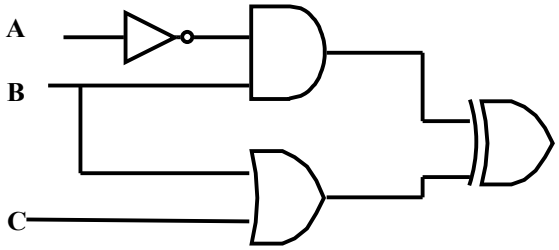
12. **03-04 C4 Digital Electronics**

List all ordered triples that make the following circuit FALSE:



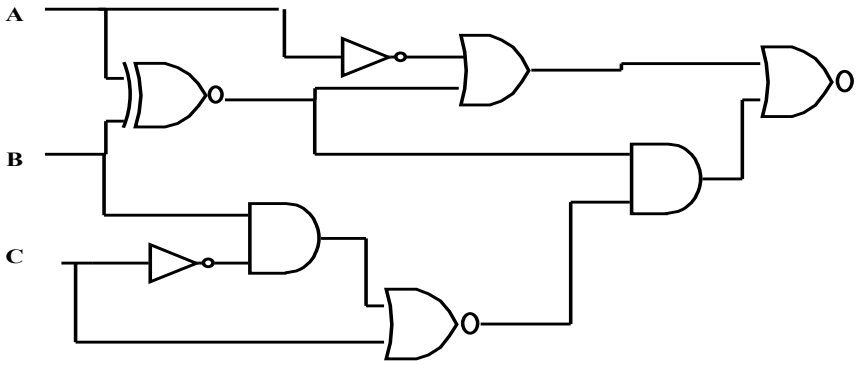
13. **04-05 C3 Digital Electronics**

How many ordered triples make the following circuit TRUE?



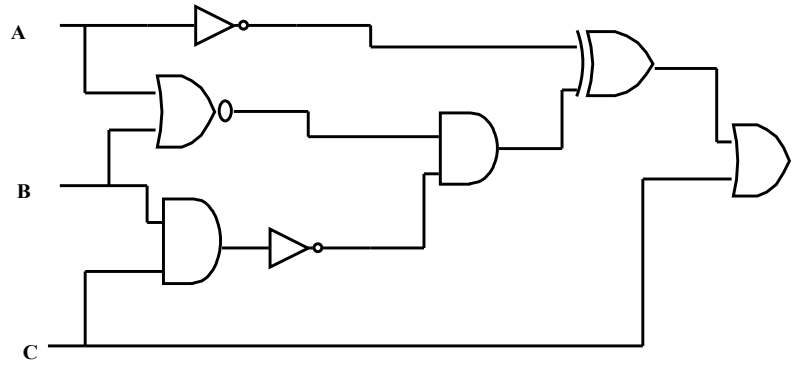
14. **04-05 C3 Digital Electronics**

Translate the following circuit to a Boolean expression. Do not simplify. Use the minimum number of parentheses.



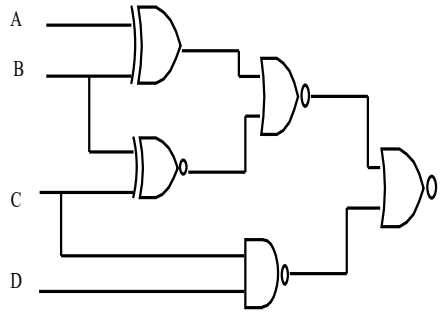
15. **04-05 C4 Digital Electronics**

Which ordered triples make the following circuit FALSE?



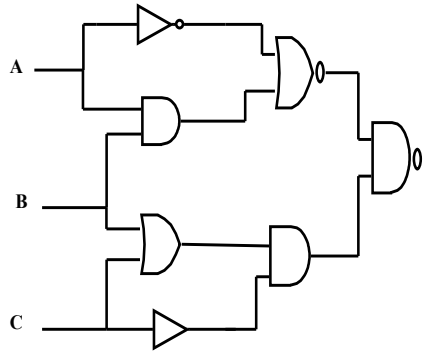
16. **05-06 C3 Digital Electronics**

How many ordered quadruples make the circuit TRUE?



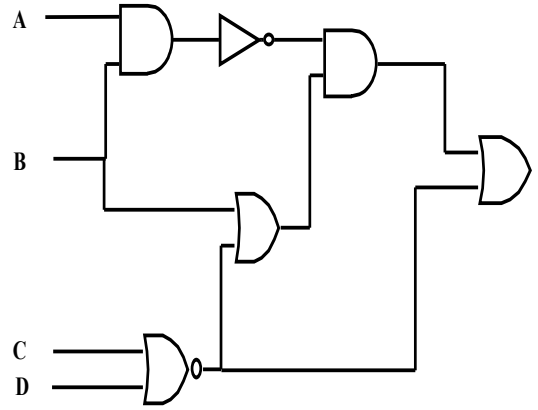
17. **05-06 C3 Digital Electronics**

Translate the following circuit to a Boolean expression.
Use the fewest sets of parentheses possible. Do not simplify.



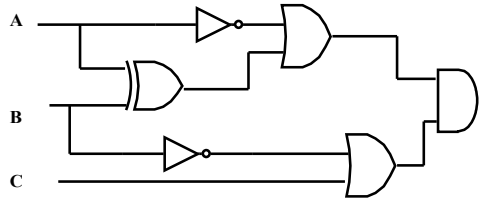
18. **05-06 C4 Digital Electronics**

How many ordered quadruplets make the following circuit TRUE?



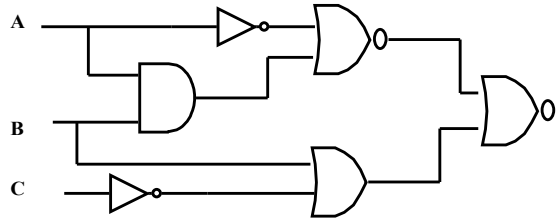
19. **06-07 C3 Digital Electronics**

How many ordered triples make the following circuit TRUE?



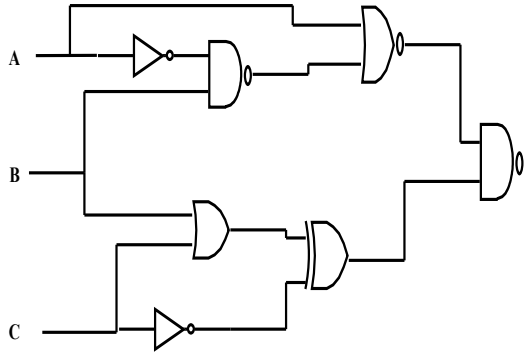
20. **06-07 C3 Digital Electronics**

Convert the following circuit to a Boolean expression and simplify.



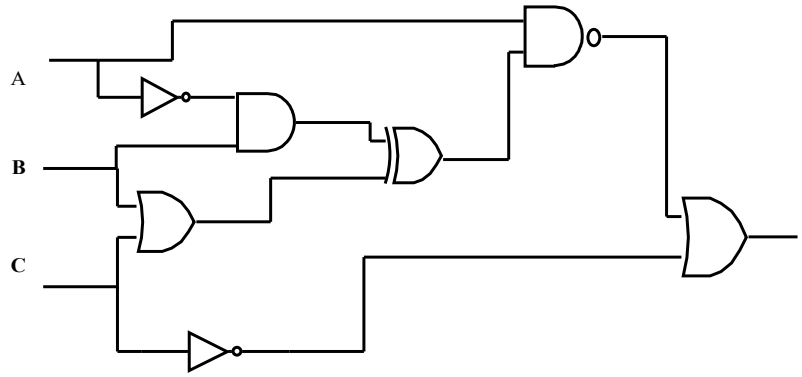
21. **06-07 C4 Digital Electronics**

List all the ordered triples that make the following circuit FALSE?



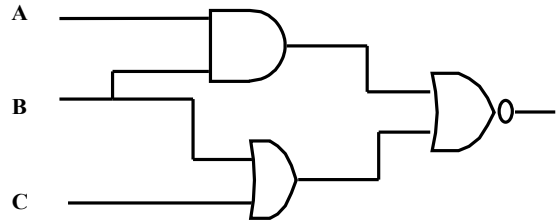
22. **07-08 C3 Digital Electronics**

How many ordered triples make the following circuit TRUE?



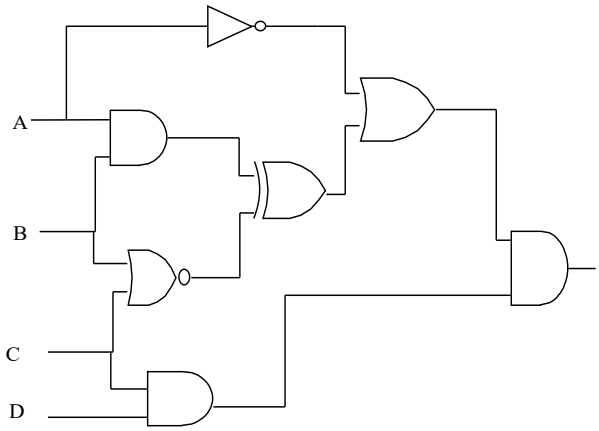
23. **07-08 C3 Digital Electronics**

List all the ordered triples that make the circuit TRUE?



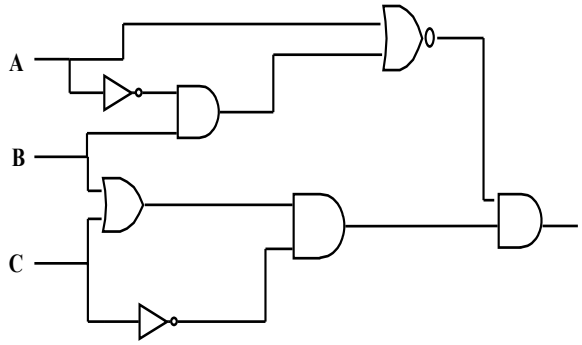
24. **07-08 C4 Digital Electronics**

List all the ordered quadruples that make the following circuit TRUE?



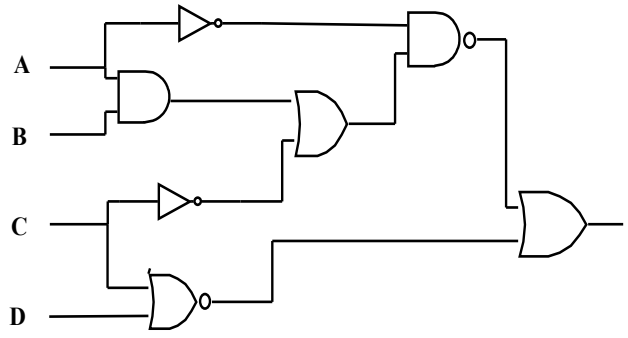
25. **08-09 C3 Digital Electronics**

Simplify the Boolean expression represented by this circuit.



26. **08-09 C3 Digital Electronics**

List all the ordered quadruples that make the circuit FALSE?

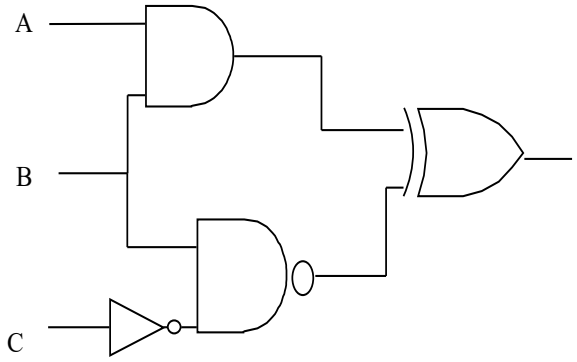


27. **08-09 C4 Digital Electronics**

List (by letter) all of the following expressions that are equivalent to this circuit.

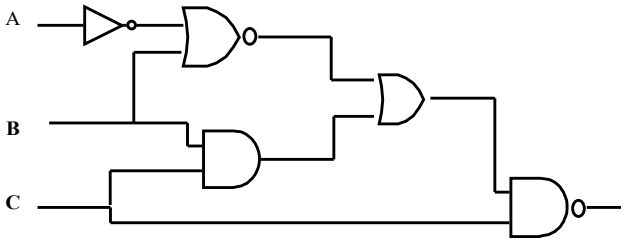
A. $\overline{BC}$ B. $AB \oplus \overline{B\overline{C}}$ C. $AB\overline{C} + \overline{B} + \overline{A}C$

D. $\overline{AB \oplus B\overline{C}}$ E. $A \oplus C$



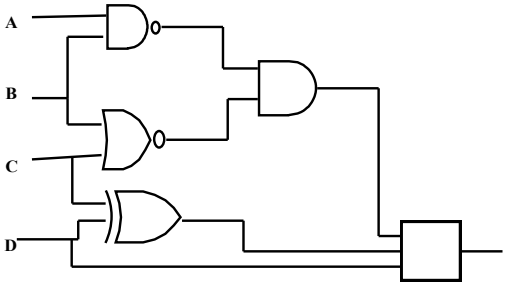
28. **09-10 C3 Digital Electronics**

How many ordered triples make the following circuit TRUE?



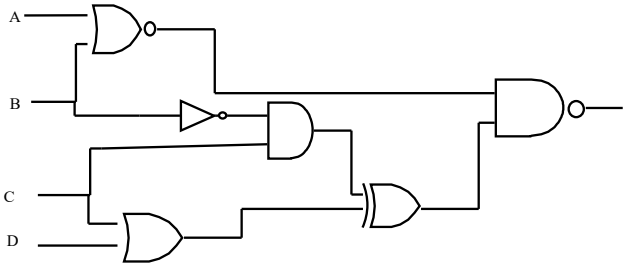
29. **09-10 C3 Digital Electronics**

Define a new gate, square, to receive 3 inputs. It is TRUE if at least 2 inputs are TRUE. How many quadruples make this circuit TRUE?



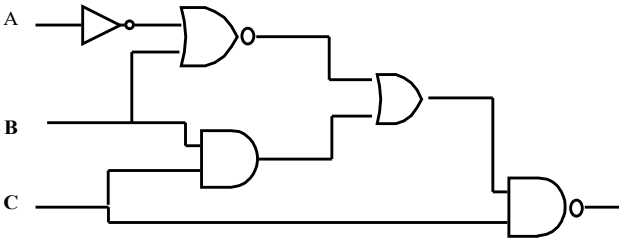
30. **09-10 C4 Digital Electronics**

How many ordered quadruples make the following circuit FALSE?



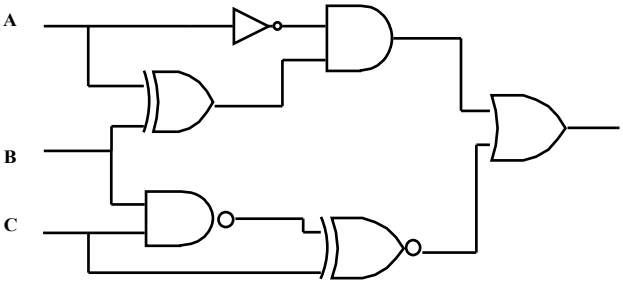
31. **10-11 C3 Digital Electronics**

How many ordered triples make this circuit TRUE?



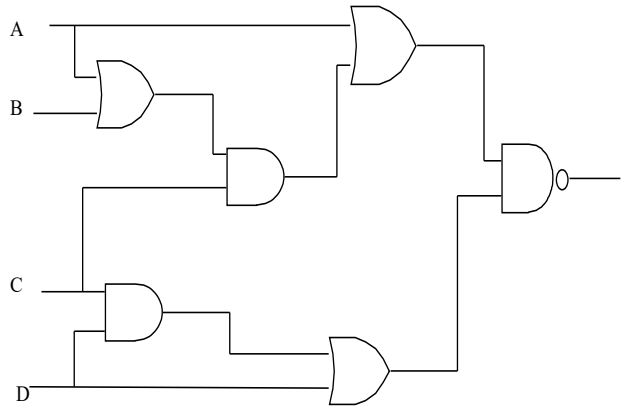
32. **10-11 C3 Digital Electronics**

Simplify the Boolean expression that is represented by this circuit.



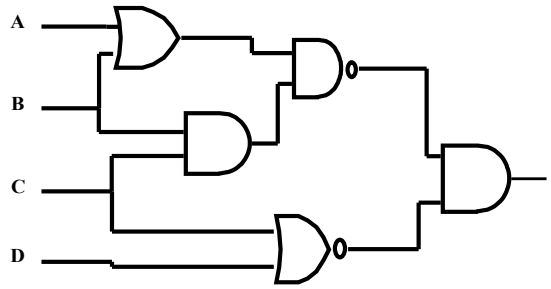
33. **10-11 C4 Digital Electronics**

How many quadruples make the digital circuit FALSE?



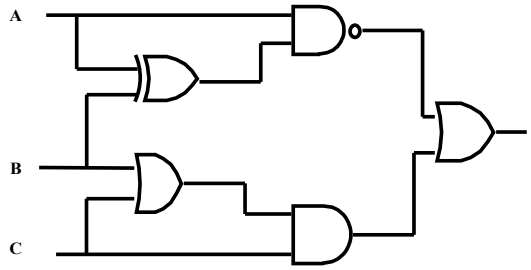
34. **11-12 C4 Digital Electronics**

Simplify the Boolean expression that this circuit represents.



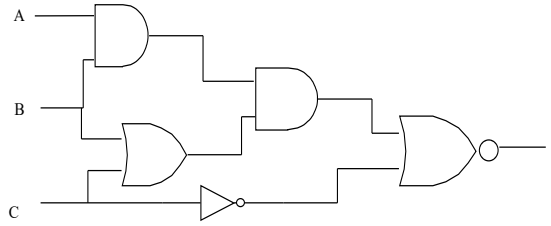
35. **11-12 C4 Digital Electronics**

Which ordered triples make this digital circuit FALSE.



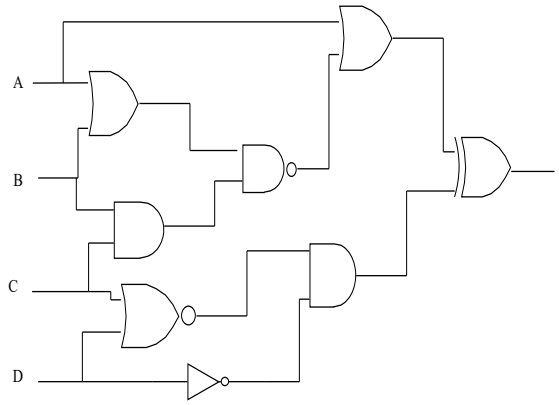
36. **12-13 C4 Digital Electronics**

List all the ordered triple(s) that make the following circuit TRUE?



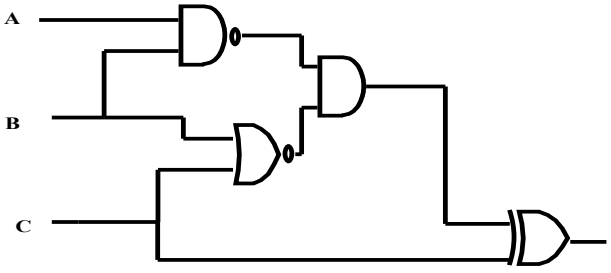
37. **12-13 C4 Digital Electronics**

How many ordered quadruples make the following circuit FALSE?



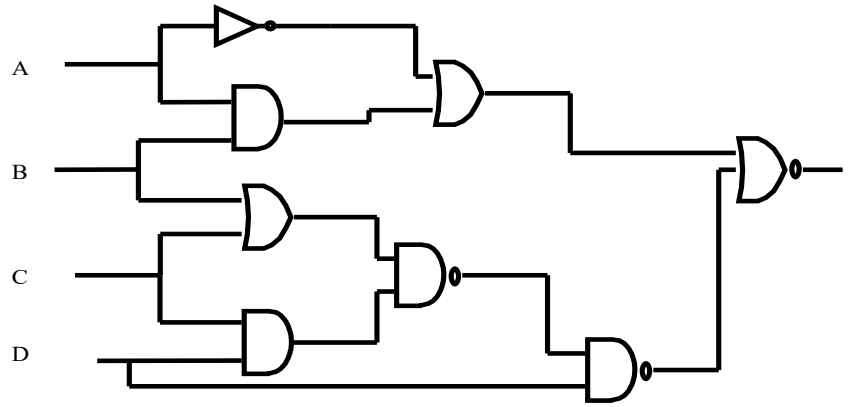
38. **13-14 C4 Digital Electronics**

Which ordered triple(s) make the following circuit true?



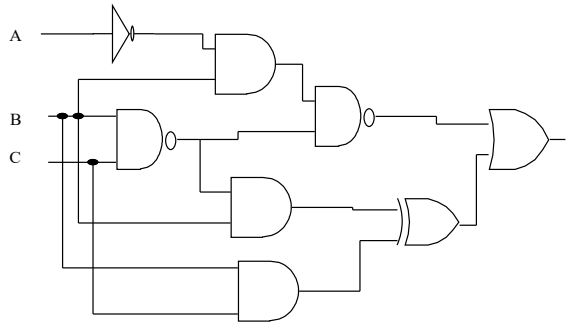
39. **13-14 C4 Digital Electronics**

Simplify the Boolean expression represented by the digital circuit.



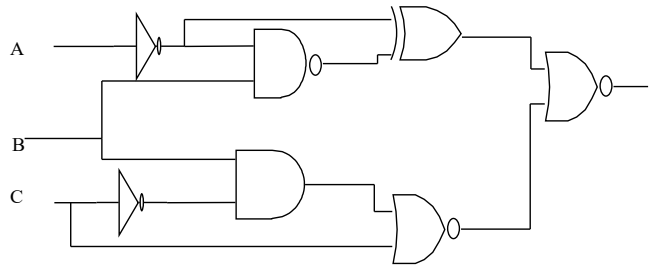
40. **14-15 C4 Digital Electronics**

Translate the circuit into a simplified Boolean expression.



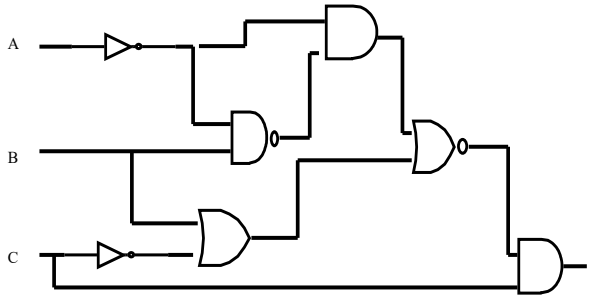
41. **14-15 C4 Digital Electronics**

Which ordered triples make the circuit TRUE?



42. **15-16 C4 Digital Electronics**

Simplify the Boolean expression produced by the digital circuit:



43. **15-16 C4 Digital Electronics**

How many ordered quadruples make the following circuit TRUE**?**

